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Experiment 1

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Experiment 2

The fluorescence of cells induced to express various combinations of encapsulin proteins, full-length Venus, and the N-terminal and C-terminal components of Venus were compared. Figure 1 shows the mean cellular fluorescence levels (bars indicate standard error of the mean).

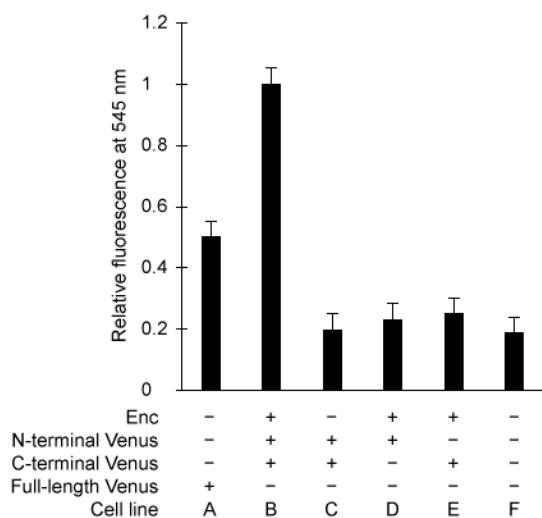


Figure 1 Effects of induced expression of encapsulin (Enc) proteins, Venus protein, and Venus protein components on the fluorescence of the eukaryotic cells in which they were expressed

Which characteristic below distinguishes viruses from encapsulins? Only viruses possess:

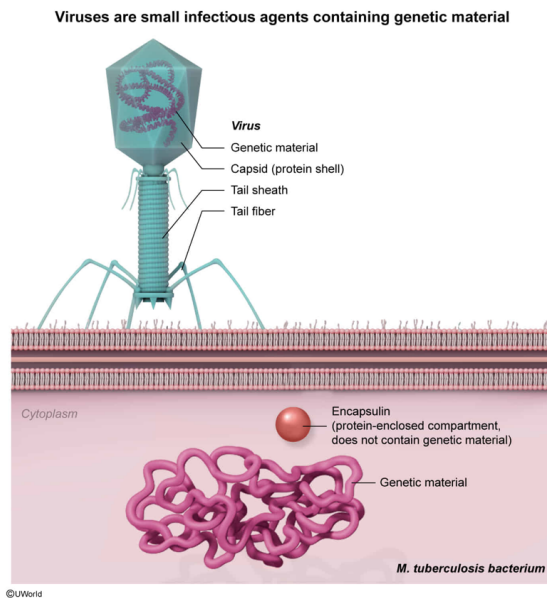
- ☐ A. chromosomes. [6%]

- ☐ B. a protein coating. [27%]
- ✓ ☒ C. genetic material. [57%]
- ☐ D. self-assembling subunits. [7%]

Correct

57% answered correctly

Explanation:



All **living cells** are assembled using instructions contained in **genomic DNA**. Genes in prokaryotes (which lack a nucleus) are located in the cytosol, often as a single, circular **chromosome**. Eukaryotic genes are found in nuclear and mitochondrial chromosomes.

Viruses are **small infectious agents containing genetic material** (**RNA or DNA**) that can **only be replicated within a living host cell**; for this reason, viruses are often described as "obligate intracellular parasites." Unlike living cells, viruses package their genomes into protein shells called capsids for reproduction.

The passage describes encapsulins as tiny, protein-enclosed compartments found in some bacterial cells, and the question asks for a characteristic distinguishing encapsulins from viruses. According to the passage, both encapsulins and viruses are protein-enclosed and self-assembling. Genetic material is a defining component of viruses, but the passage includes no information to suggest that encapsulins contain genetic material. Therefore, the presence of **genetic material distinguishes viruses from encapsulins**.

(Choice A) Neither viruses nor encapsulins contain chromosomes.

(Choices B and D) Both viruses and encapsulins are enclosed in protein coats that self-assemble; therefore, these characteristics do not distinguish viruses from encapsulins.

Educational objective:

Viruses are small, protein-enclosed infectious agents containing genetic material, and can only be replicated within a host cell.

Time spent:0 sec

Subject:

Biology

QID:403625

System:

Cellular Biology

Some prokaryotic cells contain encapsulins, which are tiny compartments self-assembled from multiple copies of a protein. This protein resembles those found in self-assembling viral capsids. Researchers studying cellular compartmentalization carried out an experiment investigating the effects of encapsulins in *Mycobacterium tuberculosis* (TB) cells, which are less sensitive to the harsh environment of phagocytic immune cells (ie, low pH, oxidants such as hydrogen peroxide) than are many other bacteria. In an additional experiment, the researchers introduced encapsulin proteins into eukaryotic cells along with the N- and C-terminal halves of a fluorescent protein called Venus. As in translated full-length Venus proteins, the assembled N- and C-terminal components have been shown to fluoresce at 545 nm when irradiated by 515 nm light.

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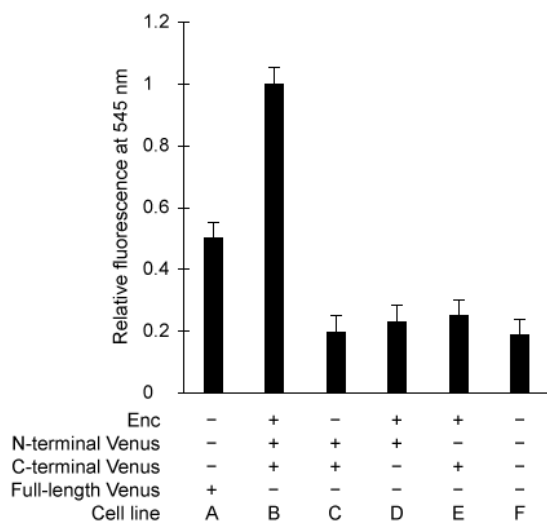


Figure 1 Effects of induced expression of encapsulin (Enc) proteins, Venus protein, and Venus protein components on the fluorescence of the eukaryotic cells in which they were expressed

Which experimental approach(es) could be used to study the effect of encapsulins on the sensitivity of TB cells to an antibiotic? Comparing:

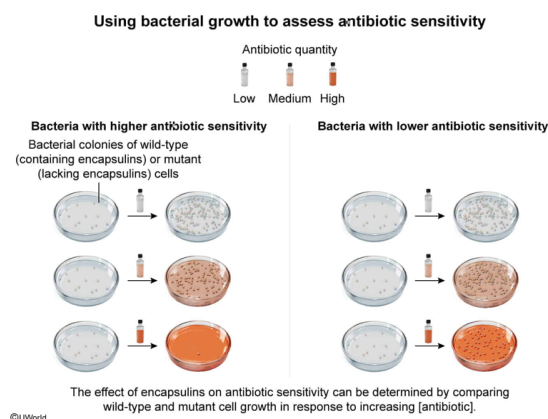
- I. the minimum concentration of antibiotic that inhibits growth of wild-type versus mutant TB cells
- II. the mean concentration of peroxides produced by wild-type versus mutant TB cells during antibiotic treatment
- III. the concentration of damaging molecules produced by immune cells engulfing wild-type versus mutant TB cells

- ✓ ☒ A. I only [41%]
- ☐ B. II only [6%]
- ☐ C. I and III only [29%]
- ☐ D. I, II, and III only [22%]

Correct

39% answered correctly

Explanation:



Certain immune cells (eg, **macrophages**) engulf pathogens inside a vesicle in a process known as **phagocytosis**. When such a vesicle fuses with a **lysosome**, a digestive **phagolysosome** is formed. Within the phagolysosome is an acidic environment (low pH) containing enzymes that break down engulfed molecules and produce damaging **reactive oxygen species** that help kill or destroy pathogens.

Some bacteria are resistant to the body's immune defenses but can be targeted by drugs (ie, antibiotics) that compromise cellular integrity (eg, by inhibiting cell wall synthesis) or the reproductive capability of bacteria. However, the combined effects of the immune response and an antibiotic are better withstood by some bacteria than by others. Continued growth and reproduction of resistant bacteria reduces the sensitivity of the total bacterial population to an antibiotic. In such cases, a higher antibiotic concentration (or a different type of antibiotic) is required to treat the infection.

The passage describes experiments in which wild-type TB cells possessing an operon

containing encapsulin and peroxidase genes were studied along with mutant TB cells lacking this operon. This question asks which experimental approaches could be used to

study the effect of encapsulins on the sensitivity of these cells to an antibiotic. **Antibiotic sensitivity relates a bacterial response** (eg, death, diminished growth) **to antibiotic concentration**. Therefore, the antibiotic sensitivity of wild-type versus mutant TB cells could be assessed by **comparing the minimum antibiotic concentrations that would inhibit their growth (Number I)**.

(Number II) The passage describes peroxides as molecules produced by *immune* cells (not TB cells) to *destroy* invading pathogens.

(Number III) Although damaging molecules produced by immune cells can contribute to *immune* defenses against TB cells, the concentrations of damaging molecules alone provide no information relating bacterial characteristics (eg, growth) to *antibiotic* sensitivity.

Educational objective:

Antibiotics are molecules that function to kill or inhibit the growth of bacterial cells. Antibiotic resistance is a state in which a higher antibiotic concentration (or a different type of antibiotic) is required to treat a bacterial infection.

Time spent:0 sec

Subject:
Biology

QID:403626

System:
Cellular Biology

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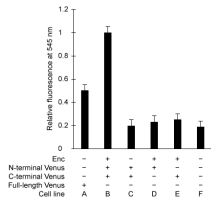
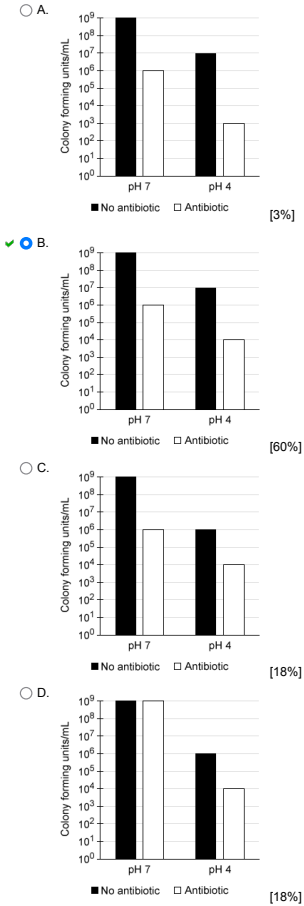


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If the hypothesis that pH has no effect on the percentage of wild-type TB cells killed by a certain antibiotic is true, which graph shows the expected amounts of wild-type TB cells?



Correct
59% answered correctly

Explanation:

pH does not affect the percent response to antibiotic.
On the x-axis with a log scale, equal unit differences represent equal ratios of values, regardless of overall magnitude.

Antibiotic-treated (log units)	Antibiotic-treated (CFU/mL)	Percent response to CFU/mL
7	10 ^{6.5}	99.9%
8	10 ^{7.5}	99.9%
9	10 ^{8.5}	99.9%
10	10 ^{9.5}	99.9%

Choice B
3-unit difference: 10^{6.5} to 10^{9.5} (100-fold decrease in CFU/mL)
3-unit difference: 10^{7.5} to 10^{10.5} (100-fold decrease in CFU/mL)
3-unit difference: 10^{8.5} to 10^{11.5} (100-fold decrease in CFU/mL)
3-unit difference: 10^{9.5} to 10^{12.5} (100-fold decrease in CFU/mL)

Because the antibiotic-induced response is 100-fold, at pH 7 and pH 4 were the same number of log units, the fold-decrease is expected.

The **null hypothesis** in an experiment states the expected result if the **independent variable** under study has **no effect on the dependent variable**. For example, if pH has no effect on the percentage of cells killed by an antibiotic, then the antibiotic-induced reduction in bacterial cell numbers should be *similar* at all pH values studied.

The question asks for the graph depicting the expected amounts of wild-type TB cells if the null hypothesis that pH has no effect on the percentage of cells killed by a certain antibiotic is true. The graphs provided depict the CFU per milliliter (a commonly used indicator of bacterial amounts) at pH 7 and at pH 4, plotted with a logarithmic y-axis.

On a **logarithmic scale**, each unit difference represents a **ten-fold** (ie, an order of magnitude) **relative difference**. Therefore, a **three-unit difference** between untreated and treated TB cell amounts at pH 7 and a three-unit difference between untreated and treated TB cell amounts at pH 4 represents the **same relative difference** (ie, $10 \times 10 \times 10 = 1,000$ -fold, or a **99.9% reduction**) between treated and untreated TB cells.

(Choice A) This graph depicts a difference of three logarithmic units at pH 7 but a difference of four logarithmic units at pH 4.

(Choice C) Although the size of the antibiotic-treated bar is 33% lower at each pH on this graph, at pH 7 the difference is three logarithmic units whereas at pH 4 the difference is two logarithmic units.

(Choice D) This graph depicts equal antibiotic-induced decrements in TB cell amounts at pH 7 and at pH 4, representing a 0.1% decrement at pH 7 and a 99% decrement at pH 4.

Educational objective:

The null hypothesis in an experiment states the expected result if the independent variable or variables under study have no effect on the dependent variable. On a logarithmic scale, each unit difference represents a ten-fold change in value.

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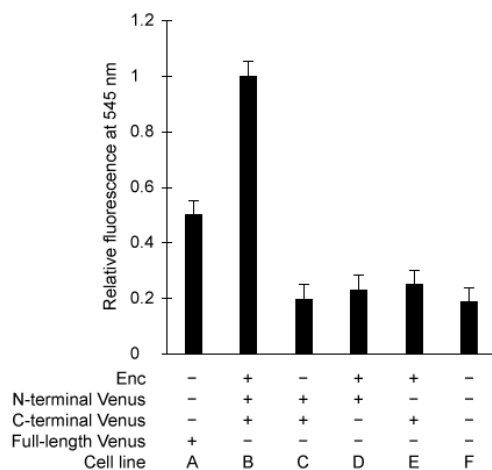


Figure 1 Effects of induced expression of encapsulin (Enc) proteins, Venus protein, and Venus protein components on the fluorescence of the eukaryotic cells in which they were expressed

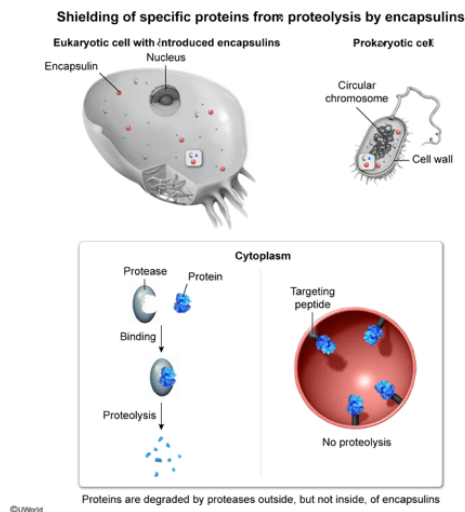
If the function of encapsulins in eukaryotic cells is the same as in bacterial cells, which of the roles below could encapsulins play in bacterial cells?

- ☐ A. Separating DNA replication proteins within a membrane-bound compartment [9%]
- ☐ B. Creating an acidic environment that contains lysosomal enzymes [11%]
- ☐ C. Enclosing specific enzymes within the mitochondrial matrix [5%]
- ☒ D. Providing a specialized compartment protecting specific proteins from proteolysis [73%]

Correct

72% answered correctly

Explanation:



Cells are defined and **classified** by their organization. All cells possess a **plasma membrane** separating intracellular components from the extracellular environment, and cells typically possess some form of **intracellular organization**. The means and extent of intracellular organization vary between **biological domains**. With few exceptions (eg, **red blood cells**), eukaryotic cells possess membrane-enclosed nuclei and other **organelles**. In contrast, prokaryotic cells (bacteria and archaea) do not contain membrane-enclosed organelles.

The passage describes encapsulins, which are very small *protein*-enclosed compartments within some prokaryotic cells, as well as an experiment in which encapsulins were experimentally introduced into eukaryotic cells. The question asks for a role encapsulins could play in bacterial (ie, prokaryotic) cells.

In bacteria, which lack the *membrane*-bound intracellular **compartmentalization** of eukaryotic cells, a protein-enclosed region such as that defined by an **encapsulin could provide a compartment protecting specific proteins from cytosolic proteases** (ie, enzymes catalyzing proteolysis, the breakdown of proteins). This protection prevents proteins within the encapsulins from being subjected to proteolysis.

Because prokaryotic cells lack membrane-bound organelles such as nuclei (**Choice A**), **lysosomes** (**Choice B**), and mitochondria (**Choice C**), encapsulins *cannot* serve a role localized to these structures in bacteria.

Educational objective:

Prokaryotic cells exhibit intracellular organization but lack specialized membrane-enclosed regions (ie, organelles).

Time spent: 0 sec
Subject:
Biology

QID: 403628
System:
Cellular Biology

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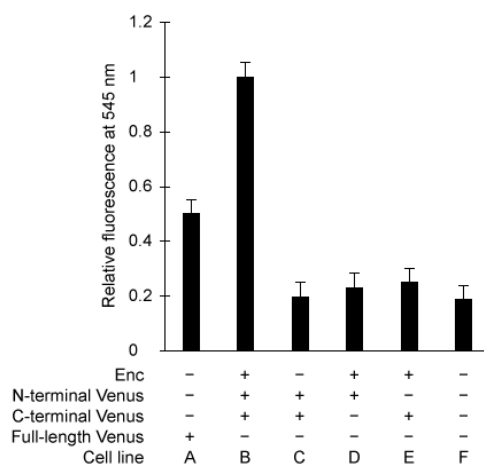
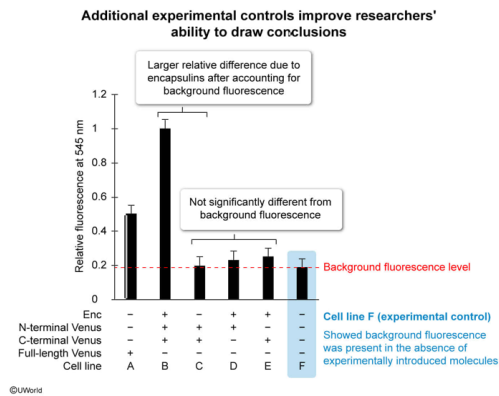


Figure 1 Effects of induced expression of encapsulin (Enc) proteins, Venus protein, and Venus protein components on the fluorescence of the eukaryotic cells in which they were expressed

Why was fluorescence measured in cell line F?

- ☐ A. This measurement is used along with the other measurements to determine encapsulin fluorescence. [3%]
- ☒ B. This measurement is a control to account for the background fluorescence level of the cells. [78%]
- ☐ C. This measurement is used to demonstrate that the cells do not express endogenous Venus. [3%]
- ☐ D. This measurement is a control to verify that the cells are capable of fluorescing. [14%]

Explanation:

Experiments often involve comparisons between two or more **groups**, with one considered the **control group** (eg, wild-type, placebo) and the other group or groups considered the **experimental** or **intervention groups** (eg, mutant, treated).

In addition to the measurements used to compare the control group and the experimental group, additional control measurements are often made to strengthen researchers' ability to draw conclusions based on their experiment. For example, "background" measurements may establish the baseline value of a variable in the setting of the experiment due to factors other than those being experimentally manipulated.

The passage describes an experiment in which researchers measured the fluorescence of six eukaryotic cell lines. Measuring the **background fluorescence** in **cell line F** (ie, cells not induced to express any of the variable proteins or protein components) would **allow** the researchers to **better estimate** the true **effects of encapsulins**, Venus, and Venus components. In this way, the researchers can **appreciate** that **Venus** is **not** the **only** cellular **component contributing to** the measured **fluorescence**.

(Choice A) In addition to the cells measured in the experiment described in the passage, cells containing encapsulins but neither full-length Venus nor Venus components would be necessary to measure potential encapsulin fluorescence. However, these cells were not studied.

(Choice C) Fluorescence emitted from cell line F would be consistent with the expression of endogenous Venus but would not prove the presence of Venus (as the fluorescence could arise from molecules other than Venus). Assessing Venus protein levels directly (eg, by **western blotting**) would be a better way of assessing whether endogenous Venus expression occurred.

(Choice D) The passage states that the assembled Venus components would fluoresce based on prior research, and the cells induced to express full-length Venus (cell line A) would serve as a positive control to demonstrate that fluorescence can be measured from the cells. Although the fluorescence measured in these cells does demonstrate the cells are capable of fluorescing, this is likely not the reason they were included in the experiment.

Educational objective:

Experimental control measurements improve researchers' ability to draw conclusions from an experiment.

Time spent: 0 sec
Subject:
Biology

QID: 403629
System:
Cellular Biology

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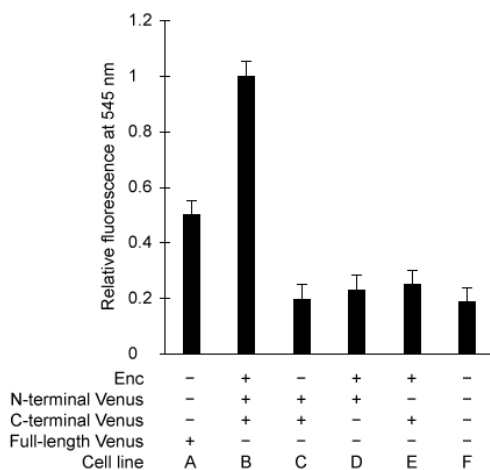


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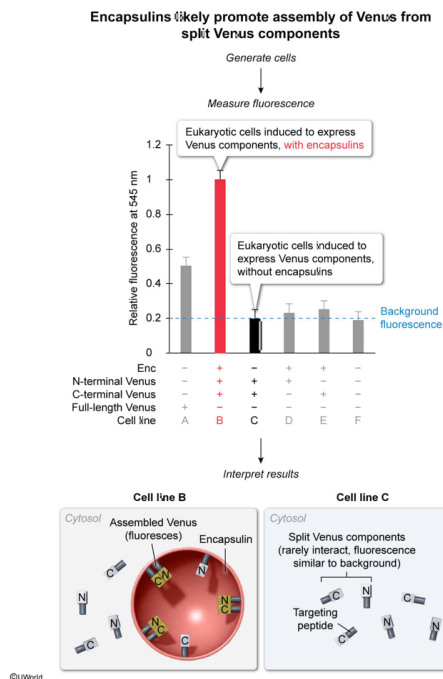
The data in Figure 1 best support which statement?

- ✓ ☒ A. Encapsulins promote assembly of Venus from the split N-Venus and C-Venus components. [52%]
- ☐ B. Translated full-length Venus emits more fluorescence than Venus comprised of assembled N- and C-Venus components. [26%]
- ☐ C. Encapsulins allow more efficient Venus production by separating its transcription and translation, as in bacteria. [11%]
- ☐ D. Translated full-length Venus is a less ideal substrate for cytosolic proteases than is assembled full-length Venus. [9%]

Correct

52% answered correctly

Explanation:



Protein synthesis culminates a **process** started when the gene for a protein is **transcribed** into an mRNA molecule. This mRNA is translated (after **processing**, in eukaryotes) by ribosomes into amino acids linked by peptide bonds, thereby forming the protein.

In eukaryotes, the transcription of a gene into an mRNA molecule and the translation of that mRNA into a protein cannot occur simultaneously, as the nuclear membrane separates the genomic DNA from the ribosomes. However, **in prokaryotes transcription and translation can occur simultaneously** because no nuclear membrane is present.

The passage describes an experiment in which the fluorescence of cells containing various combinations of encapsulin proteins and the N-terminal and C-terminal components of Venus were compared. The results presented in Figure 1 show that the fluorescence of cells containing both N- and C-Venus components was dramatically higher when encapsulins were also present (ie, cell line B) than when encapsulins were absent (ie, cell line C). These **findings support** the possibility that **encapsulins promote the assembly of Venus** from the split N-Venus and C-Venus components.

(Choices B and D) Both the **intrinsic fluorescence** and the **proteolytic susceptibility** of translated full-length Venus versus Venus assembled from components would be better addressed by studying cells with similar amounts of full-length or assembled Venus in the absence of encapsulins. Nevertheless, the *lower* fluorescence of cells with full-length Venus (ie, cell line A) than cells with encapsulins and the split Venus components (ie, cell line B) argues against both possibilities.

(Choice C) An encapsulin-induced increase in Venus production efficiency is consistent with the high fluorescence observed in the cells with both encapsulins and the split Venus components (ie, cell line B). However, transcription and translation are not separated in bacteria (ie, both occur in the cytosol).

Educational objective:

Lacking membrane-enclosed organelles like nuclei, bacteria do not separate the processes of transcription and translation. However, cellular processes may be facilitated in bacteria by other forms of cellular organization.

Time spent:0 sec

Subject:

Biology

QID:403630

System:

Cellular Biology